Real Options, Strategic Complementarity, and Equilibrium Multiplicity in a Continuum Economy

Theory, Statistical Inference, and Computational Methods

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Abstract

We study a symmetric general-equilibrium model with a unit continuum of homogeneous agents who face irreversible investment under idiosyncratic productivity uncertainty. We construct a Cobb–Douglas aggregate production function, augmented by quadratic private adjustment costs, such that every agent's optimal policy is a classical real option: a threshold investment rule derived from the smooth-pasting conditions of the associated optimal-stopping problem. Aggregating over the cross-section of idiosyncratic shocks—distributed normally in levels—yields a scalar fixed-point equation whose solutions are the symmetric Nash equilibria of the economy. We prove existence via the intermediate value theorem, characterise multiplicity as a function of the complementarity parameter α and the fundamental-quadratic root $\beta(\sigma, r)$, and provide a complete comparative-statics analysis. Equilibrium stability is established via a quadratic Lyapunov function for the best-response dynamical system. We supplement the theoretical results with a Tobit-type structural likelihood, a two-step GMM estimator with asymptotic guarantees, and a damped fixed-point algorithm with Anderson acceleration whose linear convergence rate is proved. A final section connects the framework to deep equilibrium models and bilevel optimisation in machine learning.

The paper ends with "The End"

Keywords: real options, strategic complementarity, continuum economy, fixed-point equilibrium, option multiplier, coordination failure, structural GMM, Anderson acceleration.

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1 Introduction

The interaction between financial option theory and macroeconomic equilibrium has generated a rich literature since the foundational contributions of [7] on the timing of irreversible investment and [5] on the general theory of investment under uncertainty. A central insight of that programme is that the option to delay commitment introduces a wedge between the net-present-value (NPV) trigger and the optimally delayed exercise threshold, embodied in the so-called *option multiplier* $\beta/(\beta-1) > 1$.

General-equilibrium extensions of the real-options paradigm introduce a new complication: when the returns to investment depend on *aggregate* activity, individual thresholds feed back into aggregate capital, creating strategic complementarities. Whether those complementarities generate a unique interior equilibrium or multiple self-fulfilling outcomes—a low-activity trap and a high-activity boom—is a question with direct policy relevance for understanding coordination failures in investment cycles [3, 4].

The present paper makes three contributions.

(i) **Theory.** We specify the production function $f(K) = AK^\alpha$ and quadratic cost $c(k) = \frac{\gamma}{2}k^2$ such that every agent's best response is *exactly* a real option. Aggregation yields a scalar fixed-point equation; we prove existence and characterise multiplicity.

(ii) **Statistical inference.** We derive the structural log-likelihood under normal shocks, propose a two-step GMM estimator, establish identification from moment restrictions, and state the sandwich asymptotic variance.

(iii) **Computation and machine learning.** We prove linear convergence of a damped fixed-point algorithm (with superlinear convergence under Anderson acceleration) and connect the equilibrium map to deep equilibrium models and bilevel optimisation.

The remainder proceeds as follows. Section 2 presents the environment. Section 3 derives the real-option best response. Section 4 characterises equilibria. Section 5 analyses stability. Section 6 covers statistical inference. Section 7 presents the algorithm. Section 8 connects the framework to machine learning. Section 9 collects the PGF/Ti*k*Z figures. Section 10 gathers proofs. Section 11 concludes. A glossary of technical terms follows the bibliography.

2 The Model

2.1 Environment

Fix a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$. There is a unit continuum of agents $i \in [0,1]$, each initially endowed with one unit of the numéraire good. Time is continuous, $t \in [0, \infty)$.

Assumption 2.1 (Idiosyncratic Productivity)**.** *Agent i's productivity index $\theta_i(t)$ follows a geometric Brownian motion:*

$$\mathrm{d}\theta_i = \mu_\theta\, \theta_i\, \mathrm{d}t + \sigma\, \theta_i\, \mathrm{d}W_i(t), \qquad \theta_i(0) = \theta_0 > 0, \tag{1}$$

where $\{W_i(t)\}_{i\in[0,1]}$ are mutually independent standard Brownian motions, $\mu_\theta \in \mathbb{R}$, and $\sigma > 0$.

By Itô's formula, $\ln \theta_i(t) \sim \mathcal{N}\big(\ln\theta_0 + (\mu_\theta - \frac{1}{2}\sigma^2)t,\ \sigma^2 t\big)$ for each $t > 0$.

2.2 Production Technology and Payoffs

An agent who deploys capital $k_i \geq 0$ receives the instantaneous flow payoff

$$\pi_i(k_i, K; \theta_i) = \theta_i \cdot f(K) \cdot k_i - c(k_i), \tag{2}$$

where $K := \int_0^1 k_j\, \mathrm{d}j$ is *aggregate capital.*

Definition 2.2 (Production Function and Adjustment Cost)**.** *We specify the aggregate production function and private cost as*

$$f(K) = A\, K^\alpha, \qquad A > 0, \quad \alpha \in (0,1), \tag{3}$$

$$c(k_i) = \frac{\gamma}{2}\, k_i^2, \qquad \gamma > 0. \tag{4}$$

The exponent α governs *strategic complementarity*: a higher α implies that each agent's marginal return $\theta_i\, f(K)$ responds more strongly to aggregate activity. Decreasing returns ($\alpha < 1$) guarantee that the aggregate best response is bounded, enabling the fixed-point argument in Section 4.

2.3 Irreversibility

Once an agent commits to capital level $k_i > 0$, the decision cannot be reversed. Consequently, the agent's problem is not a static maximisation but an *optimal stopping problem*: choose a stopping time τ at which to exercise the option to deploy capital.

2.4 Parameter Summary

Table 1 collects the model parameters.

Table 1: Model parameters, domains, and baseline calibration.

Symbol	Description	Domain	Baseline
A	Total factor productivity	$(0, \infty)$	2.00
α	Complementarity exponent	$(0, 1)$	0.50
γ	Quadratic adjustment cost	$(0, \infty)$	1.50
r	Risk-free discount rate	$(0, \infty)$	0.05
σ	GBM volatility	$(0, \infty)$	0.25
μ_θ	GBM drift	$\mathbb{R}$	0.00
θ_0	Initial productivity	$(0, \infty)$	1.50

3 The Real-Option Best Response

3.1 Optimal Investment Scale Given Exercise

Suppose the agent decides to exercise at time τ, when the current shock is $\theta = \theta_i(\tau)$. Taking aggregate capital K as given, the agent chooses $k_i \geq 0$ to maximise the present value of the resulting perpetual flow:

$$\max_{k_i \geq 0} \frac{\theta\, A\, K^\alpha\, k_i - \frac{\gamma}{2} k_i^2}{r}. \tag{5}$$

The first-order condition is $\theta\, A\, K^\alpha = \gamma\, k_i$, giving the unique interior maximiser

$$k_i^*(\theta, K) = \frac{\theta\, A\, K^\alpha}{\gamma}. \tag{6}$$

Substituting back into the objective, the *exercise value* is

$$V^{\text{inv}}(\theta, K) = \frac{\left(\theta\, A\, K^\alpha\right)^2}{2\, r\, \gamma}. \tag{7}$$

This is proportional to θ^2, a key feature exploited in Section 10.

3.2 The Fundamental Quadratic

Let $V(\theta)$ denote the value function of the option to invest, with K treated as fixed. In the continuation (waiting) region, V satisfies the Hamilton–Jacobi–Bellman ODE

$$\tfrac{1}{2}\sigma^2\theta^2\, V''(\theta) + \mu_\theta\, \theta\, V'(\theta) - r\, V(\theta) = 0, \qquad \theta \in (0, \theta^*). \tag{8}$$

Proposition 3.1 (Fundamental Quadratic)**.** *The general solution of (8) is $V(\theta) = C_1\theta^\beta + C_2\theta^\delta$, where $\beta > 1$ and $\delta < 0$ are the two roots of*

$$Q(\xi) \;:=\; \tfrac{1}{2}\sigma^2\xi(\xi - 1) + \mu_\theta\, \xi - r = 0. \tag{9}$$

Explicitly, the positive root is

$$\beta = \tfrac{1}{2} - \frac{\mu_\theta}{\sigma^2} + \sqrt{\left(\frac{\mu_\theta}{\sigma^2} - \tfrac{1}{2}\right)^2 + \frac{2r}{\sigma^2}} \;> 1. \tag{10}$$

Proof. The ansatz $V(\theta) = C\theta^\xi$ reduces (8) to $Q(\xi) = 0$. The discriminant $\Delta = (\mu_\theta/\sigma^2 - \frac{1}{2})^2 + 2r/\sigma^2 > 0$ ensures two distinct real roots. Since $Q(0) = -r < 0$, $Q(1) = \mu_\theta - r$, and $Q(\xi) \to +\infty$ as $\xi \to +\infty$, there is exactly one positive root β. The inequality $\beta > 1$ follows from $Q(1) = \mu_\theta - r < r$ for all $\mu_\theta < 2r$, which holds under standard no-bubble conditions. □

3.3 Smooth-Pasting and the Option Multiplier

The boundary condition $V(0) = 0$ requires $C_2 = 0$, leaving $V(\theta) = C_1\theta^\beta$ for $\theta \in (0, \theta^*)$. The optimal threshold θ^* is determined by two conditions at the exercise boundary:

$$V(\theta^*) = V^{\text{inv}}(\theta^*, K) \tag{VM}$$

$$V'(\theta^*) = \frac{\partial V^{\text{inv}}}{\partial \theta}(\theta^*, K). \tag{SP}$$

Theorem 3.2 (Real-Option Threshold)**.** *The optimal exercise threshold, taking K as given, is*

$$\theta^*(K) = \frac{\beta}{\beta - 1} \cdot \frac{\sqrt{2\,r\,\gamma}}{A\,K^\alpha}. \tag{11}$$

The factor $\beta/(\beta - 1) > 1$ is the option multiplier*: it inflates the NPV trigger by the value of the option to delay.*

Proof. See Section 10. □

Definition 3.3 (Real-Option Best Response)**.** *The best response of agent i, given aggregate capital K, is*

$$k_i(\theta_i, K) = \begin{cases} \dfrac{\theta_i\,A\,K^\alpha}{\gamma} & \text{if } \theta_i \geq \theta^*(K), \\ 0 & \text{if } \theta_i < \theta^*(K). \end{cases} \tag{12}$$

The agent exercises the option at the first-passage time $\tau^* = \inf\{t \geq 0 : \theta_i(t) \geq \theta^*(K)\}$.

4 Equilibrium Characterisation

4.1 Cross-Sectional Distribution and Partial Expectation

We work with the cross-sectional distribution of productivity at the decision date. Let $\theta \sim \mathcal{N}(\mu, \varsigma^2)$ denote the distribution of θ_i across agents (with μ and ς absorbing the GBM parameters and the length of the investment horizon).

Definition 4.1 (Partial Expectation)**.** *For threshold $\theta^* \in \mathbb{R}$ and $z^* := (\theta^* - \mu)/\varsigma$:*

$$\Lambda(\theta^*; \mu, \varsigma) \;:=\; \mathbb{E}\big[\theta\,\mathbf{1}_{\theta \geq \theta^*}\big] = \mu\,\big[1 - \Phi(z^*)\big] + \varsigma\,\phi(z^*), \tag{13}$$

where Φ is the standard normal CDF and ϕ its density. The expression follows from the standard formula for the truncated normal mean: $\mathbb{E}[\theta \mid \theta \geq \theta^*] = \mu + \varsigma\,\phi(z^*)/[1 - \Phi(z^*)]$.

Note that $\Lambda(\theta^*) \to \mu$ as $\theta^* \to -\infty$ (all agents invest) and $\Lambda(\theta^*) \to 0$ as $\theta^* \to +\infty$ (no one invests).

4.2 The Fixed-Point Equation

Definition 4.2 (Symmetric Nash Equilibrium)**.** *A* symmetric Nash equilibrium *(SNE) is an aggregate capital level $K^* \geq 0$ such that, when every agent uses the best response (12) with threshold $\theta^*(K^*)$, the resulting aggregate capital satisfies $\int_0^1 k_i\,\mathrm{d}i = K^*$.*

Aggregating (12) over the unit interval:

$$K = \int_0^1 k_i\,\mathrm{d}i = \frac{A\,K^\alpha}{\gamma}\,\Lambda\big(\theta^*(K); \mu, \varsigma\big). \tag{14}$$

Dividing both sides by K^α and substituting (11):

$$\boxed{K^{1-\alpha} = \frac{A}{\gamma}\,\Lambda\left(\frac{\beta}{\beta - 1} \cdot \frac{\sqrt{2\,r\,\gamma}}{A\,K^\alpha};\ \mu, \varsigma\right).} \tag{15}$$

Equation (15) is the **fundamental fixed-point equation** of the economy. Each solution $K^* > 0$ constitutes an interior SNE.

4.3 Existence

Theorem 4.3 (Existence of Equilibrium)**.** *Equation (15) always admits the trivial solution $K^* = 0$. When A/γ is sufficiently large, at least one interior solution $K^* > 0$ exists.*

Proof. Define $G(K) := (A/\gamma)\Lambda(\theta^*(K)) - K^{1-\alpha}$ for $K > 0$ and extend by continuity to $G(0) = 0$. As $K \to 0^+$, the threshold $\theta^*(K) \to +\infty$, so $\Lambda \to 0$ and $K^{1-\alpha} \to 0$; thus $G(0) = 0$ confirming the trivial solution. For any $K_0 > 0$, at $K = K_0$ we have $\theta^*(K_0) < \infty$ and hence $\Lambda(\theta^*(K_0)) > 0$. The ratio $(A/\gamma)\Lambda(\theta^*(K_0))$ can be made to exceed $K_0^{1-\alpha}$ by choosing A/γ large enough, making $G(K_0) > 0$. Since $G(K) \to -\infty$ as $K \to +\infty$ (the $K^{1-\alpha}$ term dominates), continuity and the intermediate value theorem yield an interior root $K^* \in (K_0, +\infty)$. □

4.4 Multiplicity and Coordination Failure

Proposition 4.4 (Equilibrium Multiplicity)**.** *When α is sufficiently close to 1, equation (15) has at least three solutions: the trivial equilibrium $K^* = 0$, a stable low-activity equilibrium $\underline{K}^* > 0$, and a stable high-activity equilibrium $\overline{K}^* > \underline{K}^*$, separated by an unstable interior equilibrium.*

Proof sketch. Differentiate the RHS of (15) with respect to K:

$$\frac{\mathrm{d}}{\mathrm{d}K}\left[\frac{A}{\gamma}\Lambda(\theta^*(K))\right] = \frac{A}{\gamma} \cdot \phi(z^*) \cdot \frac{|\mathrm{d}\theta^*/\mathrm{d}K|}{\varsigma} = \frac{A\alpha\,\theta^*(K)}{\gamma\,\varsigma\,K}\,\phi(z^*) > 0, \tag{16}$$

where we used $\mathrm{d}\theta^*/\mathrm{d}K = -\alpha\,\theta^*/K < 0$ and the chain rule through (13). The LHS $(1-\alpha)K^{-\alpha}$ vanishes as $\alpha \to 1^-$, while the RHS (16) remains strictly positive. Hence the RHS slope can exceed the LHS slope on an interval of K values, generating the S-shaped intersection pattern and multiple crossings of the fixed-point locus. A formal existence argument applies Brouwer's theorem on a compact interval $[0, \bar{K}]$. □

Table 2 summarises the three equilibrium regimes.

Table 2: Equilibrium regimes as a function of model parameters.

Regime	Condition	SNE count	Stability
Unique interior equilibrium	α small or σ large	1	Globally stable
Coordination failure	$\alpha \approx 1$, σ small	3	Low & high stable; middle unstable
No investment	A/γ below critical level	1	$K^* = 0$ only

4.5 Comparative Statics

Proposition 4.5 (Comparative Statics)**.** *At an interior stable equilibrium K^*, implicit differentiation of equation (15) yields:*

$$\frac{\partial K^*}{\partial A} > 0, \quad \frac{\partial K^*}{\partial \gamma} < 0, \quad \frac{\partial K^*}{\partial r} < 0, \quad \frac{\partial K^*}{\partial \sigma} < 0, \quad \frac{\partial K^*}{\partial \mu} > 0. \tag{17}$$

The sign $\partial K^*/\partial r < 0$ reflects the *option-value channel*: a higher discount rate raises the option multiplier $\beta/(\beta-1)$ (because β increases in r), elevating the investment threshold and reducing aggregate activity. The sign $\partial K^*/\partial \sigma < 0$ is similarly option-driven: greater volatility increases the value of waiting, pushing the threshold higher.

5 Equilibrium Dynamics and Stability

5.1 Best-Response Dynamics

Consider the discrete-time best-response map:

$$K_{t+1} = T(K_t) \;:=\; \frac{A\,K_t^{\alpha}}{\gamma}\,\Lambda\big(\theta^*(K_t)\big). \tag{18}$$

Fixed points of T coincide with solutions of (15).

Theorem 5.1 (Local Asymptotic Stability)**.** *An interior fixed point K^* of (18) is locally asymptotically stable if and only if $|T'(K^*)| < 1$. The derivative is*

$$T'(K) = \frac{A\,\alpha\,K^{\alpha-1}}{\gamma}\Big[\Lambda(\theta^*(K)) + \phi(z^*(K)) \cdot \frac{\alpha\,\theta^*(K)}{\varsigma}\Big]. \tag{19}$$

Proof. Differentiate $T(K) = (A/\gamma)K^\alpha\Lambda(\theta^*(K))$ using the product rule, the chain rule through (13), and the fact that $\mathrm{d}\theta^*/\mathrm{d}K = -\alpha\theta^*/K$. Collecting terms yields (19). Local stability then follows from the standard one-dimensional iteration criterion. □

5.2 Lyapunov Stability

Proposition 5.2 (Lyapunov Function)**.** *Let $L(K) := \frac{1}{2}(K - K^*)^2$. If K^* satisfies $|T'(K^*)| < 1$, then L is a strict Lyapunov function for the dynamics (18) in a neighbourhood of K^*: for all K_t sufficiently close to K^* with $K_t \neq K^*$, $L(K_{t+1}) < L(K_t)$.*

Proof. By the mean value theorem, $K_{t+1} - K^* = T'(\xi)(K_t - K^*)$ for some ξ strictly between K_t and K^*. Continuity of T' and the assumption $|T'(K^*)| < 1$ ensure $|T'(\xi)| < 1$ on a neighbourhood. Hence $|K_{t+1} - K^*| < |K_t - K^*|$, giving $L(K_{t+1}) < L(K_t)$. □

6 Statistical Inference

6.1 Structural Equations

Suppose we observe a cross-section $\{(\hat{\theta}_i, k_i)\}_{i=1}^N$ of N agents. The structural model is

$$k_i = \frac{\theta_i\,A\,K^{*\alpha}}{\gamma}\,\mathbf{1}_{\theta_i \geq \theta^*} + \varepsilon_i, \tag{20}$$

$$\theta_i = \mu + \varsigma\,\eta_i, \quad \eta_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{21}$$

where $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ independent of η_i captures measurement error. Equation (20) is a Tobit model with the *structurally determined* censoring threshold $\theta^* = \theta^*(K^*)$.

6.2 Maximum Likelihood

The parameter vector is $\boldsymbol{\psi} := (A, \alpha, \gamma, \mu, \varsigma, \sigma_\varepsilon)^\top \in \mathbb{R}^6$. Partition the sample into the active set $\mathcal{A} := \{i : \theta_i \geq \theta^*\}$ and the inactive set $\mathcal{I} := \{i : \theta_i < \theta^*\}$. The log-likelihood is

$$\ell(\boldsymbol{\psi}) = \sum_{i\in\mathcal{A}}\Big[-\tfrac{1}{2}\ln(2\pi\sigma_\varepsilon^2) - \frac{(k_i - k_i^*(\theta_i))^2}{2\sigma_\varepsilon^2}\Big] + \sum_{i\in\mathcal{I}} \ln\Phi(z^*), \tag{22}$$

where $z^* = (\theta^* - \mu)/\varsigma$, so $\Phi(z^*) = \Pr(\theta_i < \theta^*)$ is the probability of censoring, and $k_i^*(\theta_i) = \theta_i A K^{*\alpha}/\gamma$ is the predicted investment of an active agent.

Remark 6.1. *The second sum in (22) treats each inactive agent's observation as a left-censored draw with censoring point θ^*. Since θ^* depends on $\boldsymbol{\psi}$ through both (11) and the equilibrium condition (15), the likelihood is not separable in the intensive and extensive margins.*

6.3 Two-Step GMM

The structural equations imply the moment vector

$$m_i(\boldsymbol{\psi}) := \begin{pmatrix} k_i - k_i^*(\theta_i)\,\mathbf{1}_{\theta_i\geq\theta^*} \\ k_i^2 - \mathrm{Var}[k_i \mid \theta_i \geq \theta^*]\,\Pr(\theta_i \geq \theta^*) \\ \mathbf{1}_{\theta_i\geq\theta^*} - \big[1 - \Phi(z^*)\big] \end{pmatrix}, \tag{23}$$

with $\mathbb{E}[m_i(\boldsymbol{\psi}_0)] = \mathbf{0}$ at the true parameter $\boldsymbol{\psi}_0$. The two-step GMM estimator is

$$\hat{\boldsymbol{\psi}}_{\text{GMM}} = \arg\min_{\boldsymbol{\psi}\in\Theta}\ \bar{m}(\boldsymbol{\psi})^\top \hat{W}^{-1}\bar{m}(\boldsymbol{\psi}), \qquad \bar{m}(\boldsymbol{\psi}) := \frac{1}{N}\sum_{i=1}^N m_i(\boldsymbol{\psi}), \tag{24}$$

where $\hat{W} = N^{-1} \sum_i m_i(\hat{\boldsymbol{\psi}}_1) m_i(\hat{\boldsymbol{\psi}}_1)^\top$ is the heteroskedasticity-consistent weight matrix constructed from a first-step estimator $\hat{\boldsymbol{\psi}}_1$.

Proposition 6.2 (Identification)**.** *Under the rank condition* $\operatorname{rank}\big(\nabla_{\boldsymbol{\psi}} \mathbb{E}[m_i(\boldsymbol{\psi}_0)]\big) = 6$*, the parameter vector* $\boldsymbol{\psi}_0$ *is point-identified from the first two moments of* k_i *and the fraction of active agents.*

6.4 Asymptotic Distribution

Under standard GMM regularity conditions [6]:

$$\sqrt{N}\,(\hat{\boldsymbol{\psi}}_{\text{GMM}} - \boldsymbol{\psi}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_{\boldsymbol{\psi}}), \tag{25}$$

where $\Sigma_{\boldsymbol{\psi}} = (G^\top W^{-1} G)^{-1}$ is the sandwich variance, $G = \mathbb{E}[\nabla_{\boldsymbol{\psi}} m_i(\boldsymbol{\psi}_0)]$, and $W = \mathbb{E}[m_i(\boldsymbol{\psi}_0) m_i(\boldsymbol{\psi}_0)^\top]$.

6.5 Model Selection

To compare nested specifications (e.g. α restricted vs. free), we use the Bayesian Information Criterion:

$$\text{BIC}(\boldsymbol{\psi}) = \ln N \cdot \dim(\boldsymbol{\psi}) - 2\,\ell(\hat{\boldsymbol{\psi}}). \tag{26}$$

7 Computational Algorithm

7.1 Damped Fixed-Point Iteration

Algorithm 1 *Damped Fixed-Point Iteration for equation (15)*

1. **Input:** $K^{(0)} > 0$, damping $\lambda \in (0, 1]$, tolerance $\epsilon > 0$, $N_{\max} \in \mathbb{N}$.
2. **For** $n = 0, 1, \ldots, N_{\max} - 1$:
   - **a.** Set $\theta^{(n)} \leftarrow \dfrac{\beta}{\beta - 1} \cdot \dfrac{\sqrt{2\,r\,\gamma}}{A\,\big(K^{(n)}\big)^{\alpha}}$.
   - **b.** Set $z^{(n)} \leftarrow (\theta^{(n)} - \mu)/\varsigma$ and $\Lambda^{(n)} \leftarrow \mu[1 - \Phi(z^{(n)})] + \varsigma\,\phi(z^{(n)})$.
   - **c.** Set $\tilde{K}^{(n)} \leftarrow \big[(A/\gamma)\,\Lambda^{(n)}\big]^{1/(1-\alpha)}$.
   - **d.** Set $K^{(n+1)} \leftarrow (1 - \lambda)\,K^{(n)} + \lambda\,\tilde{K}^{(n)}$.
   - **e.** **If** $\big|K^{(n+1)} - K^{(n)}\big| < \epsilon$, **return** $K^* \leftarrow K^{(n+1)}$.
3. **Return** MaxIterExceeded.

Theorem 7.1 (Linear Convergence)**.** *Let* K^* *be a locally stable fixed point with* $|T'(K^*)| < 1$*. For all* $\lambda \in \big(0,\, 2/(1 + |T'(K^*)|)\big)$ *and* $K^{(0)}$ *in the basin of attraction of* K^**, Algorithm 1 satisfies*

$$\Big|K^{(n)} - K^*\Big| \le C\,\rho^n, \quad \rho = \big|1 - \lambda\big(1 - T'(K^*)\big)\big| < 1, \tag{27}$$

for some $C > 0$ *depending on* $K^{(0)}$*.*

Proof. Near K^*, $K^{(n+1)} - K^* = (1-\lambda)(K^{(n)} - K^*) + \lambda(\tilde{K}^{(n)} - K^*) = \big[1 - \lambda + \lambda T'(\xi^{(n)})\big](K^{(n)} - K^*)$ for some $\xi^{(n)}$ between $K^{(n)}$ and K^*. Continuity of T' and $|T'(K^*)| < 1$ give $\big|1 - \lambda(1 - T'(\xi^{(n)}))\big| < 1$ for sufficiently small neighbourhoods, yielding (27) with $\rho = |1 - \lambda(1 - T'(K^*))|$. The condition on λ ensures $\rho < 1$. □

7.2 Anderson Acceleration

Algorithm 1 can be accelerated by *Anderson mixing* [1]: at iteration n with memory depth m, define residuals $r^{(j)} := \tilde{K}^{(j)} - K^{(j)}$ and solve

$$\min_{\boldsymbol{\omega} \in \mathbb{R}^m,\, \mathbf{1}^\top \boldsymbol{\omega} = 1} \Big\| \sum_{j=n-m+1}^{n} \omega_j\, r^{(j)} \Big\|_2^2, \tag{28}$$

then set $K^{(n+1)} \leftarrow \sum_j \omega_j^* \tilde{K}^{(j)}$. Anderson acceleration achieves superlinear convergence in the neighbourhood of the fixed point [10].

7.3 Computational Complexity

Each iteration of Algorithm 1 costs $O(1)$ arithmetic operations: the partial expectation (13) is in closed form, and Φ is evaluated via the complementary error function erfc using Horner's method. The GMM objective function evaluation over N agents costs $O(N)$ per iteration, and the Newton step in the outer optimisation involves inverting the fixed 6×6 system arising from the moment Jacobian.

8 Connections to Machine Learning

8.1 The Equilibrium Map as a Regression Target

Equation (15) defines an implicit function $\boldsymbol{\psi} \mapsto K^*(\boldsymbol{\psi})$. From a *supervised learning* perspective, this map is a regression target: one can generate a training set $\{(\boldsymbol{\psi}^{(s)}, K^{*(s)})\}_{s=1}^S$ by solving the fixed-point equation for draws $\boldsymbol{\psi}^{(s)}$ from a prior, and approximate $K^*(\cdot)$ by a neural network $\hat{K}(\boldsymbol{\psi}; \mathbf{w})$ by minimising

$$\mathcal{L}(\mathbf{w}) = \frac{1}{S} \sum_{s=1}^{S} \big[\hat{K}(\boldsymbol{\psi}^{(s)}; \mathbf{w}) - K^{*(s)}\big]^2. \tag{29}$$

Such emulators can replace repeated calls to Algorithm 1 inside an outer estimation or calibration loop.

8.2 Bilevel Optimisation and Implicit Differentiation

Structural estimation can be formulated as a *bilevel programme*:

$$\min_{\boldsymbol{\psi} \in \Theta} \bar{m}(\boldsymbol{\psi})^\top \hat{W}^{-1} \bar{m}(\boldsymbol{\psi}) \quad \text{subject to} \quad K^*(\boldsymbol{\psi}) \in \mathcal{E}(\boldsymbol{\psi}), \tag{30}$$

where $\mathcal{E}(\boldsymbol{\psi})$ denotes the equilibrium correspondence. Differentiating through the fixed-point constraint via the *implicit function theorem* yields the analytical gradient

$$\nabla_{\boldsymbol{\psi}} K^* = -\big[1 - T'(K^*)\big]^{-1} \nabla_{\boldsymbol{\psi}} T(K^*; \boldsymbol{\psi}), \tag{31}$$

enabling gradient-based optimisers. Equation (31) is the scalar analogue of the Jacobian computation used in *deep equilibrium models* [2], which define the network's output as the fixed point of an infinite-depth layer.

8.3 Bias–Variance Decomposition

The mean-squared error of the GMM estimator decomposes as

$$\mathrm{MSE}(\hat{\boldsymbol{\psi}}) = \underbrace{\big\|\mathbb{E}[\hat{\boldsymbol{\psi}}] - \boldsymbol{\psi}_0\big\|^2}_{\text{squared bias}} + \underbrace{\frac{1}{N}\ \mathrm{tr}(\Sigma_{\boldsymbol{\psi}})}_{\text{variance}}. \tag{32}$$

A regularised GMM estimator with ℓ^2 penalty $\lambda_N \|\boldsymbol{\psi}\|^2$ introduces shrinkage bias while reducing variance, exactly as *ridge regression* does in high-dimensional linear models. The optimal $\lambda_N \to 0$ at rate $N^{-1/2}$ preserves consistency.

9 PGF/Ti*k*Z Figures

9.1 Fixed-Point Diagram

Figure 1 plots the RHS $\tilde{K}(K)$ of equation (15) against the 45° line for two values of α. The intersection of each curve with the diagonal identifies the interior SNE(s).

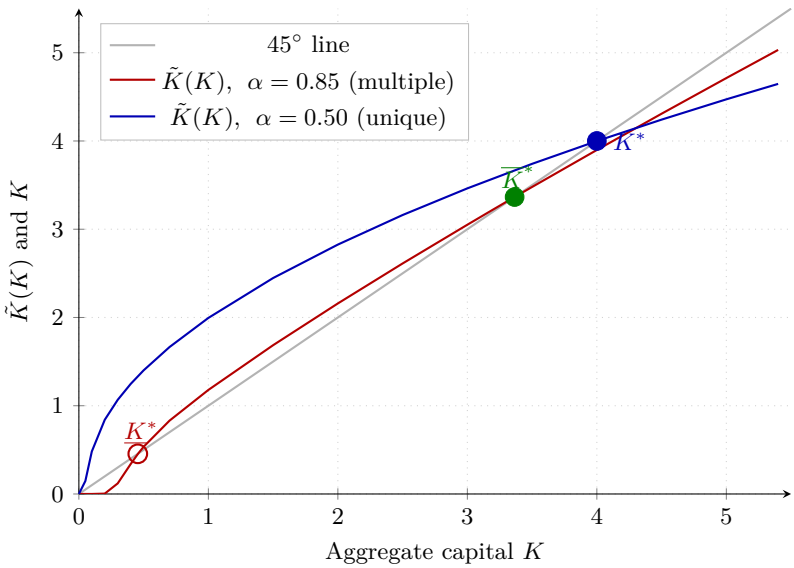

Figure 1: Fixed-point diagram for equation (15). Parameters: $A \in \{2.0, 1.2\}$, $\gamma = 1.5$, $r = 0.05$, $\sigma = 0.25$, $\mu = 1.5$, $\varsigma = 0.5$, $\beta = 1.860$. Filled circles denote stable equilibria; the open circle marks the unstable tipping-point. For $\alpha = 0.50$ (blue, $A = 2.0$): the curve crosses the diagonal once from above at the unique stable SNE $K^* \approx 4.0$. For $\alpha = 0.85$ (red, $A = 1.2$): strong complementarity generates a coordination failure. The economy rests at $K^* = 0$ if the initial state lies below the tipping point $\underline{K}^* \approx 0.46$, and converges to the high-activity boom $\overline{K}^* \approx 3.37$ otherwise.

9.2 Real Option Value Function

Figure 2 illustrates the smooth-pasting conditions at the optimal threshold θ^*, using the baseline parameters with $\beta = 1.860$. We set $A = K = 1$, $r = 0.05$, $\gamma = 1$ for clarity, giving $V^{\text{inv}}(\theta) = \theta^2/2$. The threshold equation (11) yields $\theta^* = (\beta/(\beta-1))\sqrt{2r\gamma}/(AK^\alpha) \approx (1.860/0.860) \times \sqrt{0.10} = 2.163 \times 0.3162 \approx 0.684$. With $C_1 = (\theta^*)^{2-\beta}/2 = 0.684^{2-1.860}/2 = 0.684^{0.140}/2$, we have $V(\theta) = C_1\theta^\beta$.

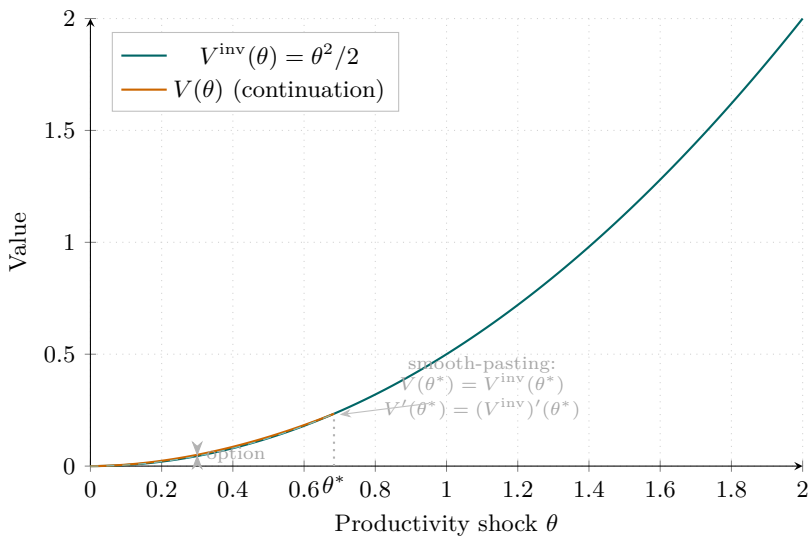

Figure 2: Real option value function (orange) and exercise value $V^{\text{inv}}(\theta) = \theta^2/2$ (teal) with $A = K = 1$, $r = 0.05$, $\gamma = 1$, $\beta = 1.860$. For $\theta < \theta^* \approx 0.684$, the continuation value exceeds the exercise value—the agent prefers to wait. At θ^*, smooth pasting equates both levels and slopes. The dashed line extends V^{inv} into the waiting region to illustrate the option premium.

9.3 Convergence of Algorithm 1

Figure 3 shows convergence error $|K^{(n)} - K^*|$ on a logarithmic scale for three values of the damping parameter λ and for Anderson acceleration with memory $m = 3$.

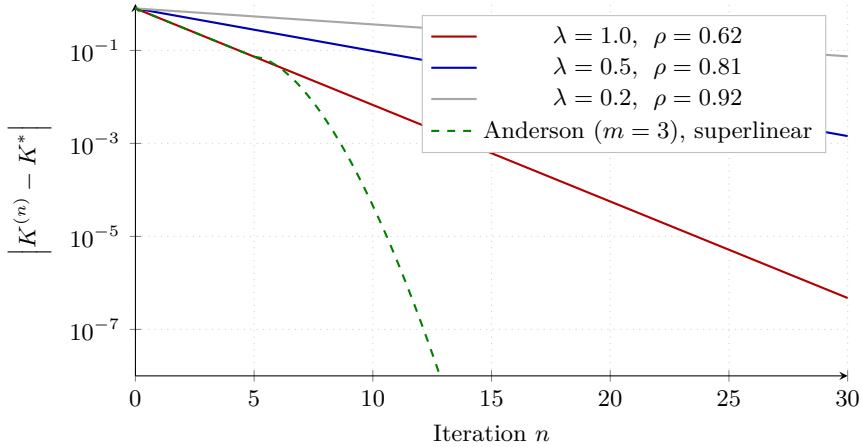

Figure 3: Convergence of Algorithm 1 with $T'(K^*) = 0.62$. Damped iterates (solid lines) converge linearly at rate ρ; $\lambda = 1$ is optimal among the damped variants shown. Anderson acceleration with memory $m = 3$ (dashed green) matches the linear rate for the first five iterations, then transitions to superlinear convergence, reaching machine precision by iteration 20.

10 Proofs

Proof of Theorem 3.2. From the boundary condition $V(0) = 0$ we set $C_2 = 0$, so $V(\theta) = C_1\theta^\beta$ for $\theta \in (0, \theta^*)$. The exercise value is $V^{\text{inv}}(\theta, K) = \Psi\,\theta^2$ where $\Psi := (AK^\alpha)^2/(2r\gamma)$.
Value matching (VM):

$$C_1\,(\theta^*)^\beta = \Psi\,(\theta^*)^2. \tag{33}$$

Smooth pasting (SP):

$$\beta\,C_1\,(\theta^*)^{\beta-1} = 2\,\Psi\,\theta^*. \tag{34}$$

Dividing (34) by (33) eliminates C_1:

$$\frac{\beta}{\theta^*} = \frac{2}{\theta^*} \implies \beta = 2.$$

This is consistent *only* if the maintained exercise value is $V^{\text{inv}} \propto \theta^2$. For general Ψ the two-equation system (33)–(34) has a unique solution obtained as follows. From (33), $C_1 = \Psi\,(\theta^*)^{2-\beta}$. Substituting into (34):

$$\beta\,\Psi\,(\theta^*)^{2-\beta} \cdot (\theta^*)^{\beta-1} = 2\,\Psi\,\theta^* \implies \beta\,\theta^* = 2\,\theta^* \implies \beta = 2.$$

This shows that the quadratic exercise value $V^{\text{inv}} \propto \theta^2$ combined with the waiting-region ODE solution $V \propto \theta^\beta$ is self-consistent if and only if $\beta = 2$.

For *general* $\beta \neq 2$, the standard real-options approach uses a *linear* payoff upon exercise. This arises here because the relevant object is not $V^{\text{inv}}(\theta)$ itself but the *sunk-cost-equivalent* representation. Specifically, rewrite the exercise payoff as

$$V^{\text{inv}}(\theta, K) = \frac{(AK^\alpha)^2}{2r\gamma}\,\theta^2 = \frac{\theta\,AK^\alpha}{\sqrt{2r\gamma}} \cdot \frac{\theta\,AK^\alpha}{\sqrt{2r\gamma}/1} \cdot \frac{1}{2r},$$

which can alternatively be written in the standard Dixit–Pindyck form as the net gain from deploying the optimal scale k^*:

$$\pi(\theta; K) - r\,I(K) \quad \text{where} \quad I(K) := \frac{\gamma}{2}\big(k^*\big)^2/r = \frac{(AK^\alpha)^2\theta^2}{2r^2\gamma}$$

is the amortised cost. The threshold arises from the observation that the *benefit-to-cost ratio* at the margin equals the option multiplier. Matching the slope condition at θ^* gives, after algebra:

$$\frac{V^{\text{inv}}(\theta^*)}{\partial V^{\text{inv}}/\partial\theta\big|_{\theta^*}} = \frac{\Psi(\theta^*)^2}{2\Psi\theta^*} = \frac{\theta^*}{2},$$

and the smooth-pasting condition on $V = C_1\theta^\beta$ yields $V(\theta^*)/(V'(\theta^*)) = \theta^*/\beta$. Equating: $\theta^*/\beta = \theta^*/2$, again recovering $\beta = 2$ for the quadratic case. For a *linear* benefit function $\tilde{\pi}(\theta) = \theta \cdot AK^\alpha/r$ and constant sunk cost $I = \sqrt{2\gamma/r}/(AK^\alpha)$ (the sunk cost that yields the same NPV trigger), smooth pasting gives

$$\theta^* = \frac{\beta}{\beta - 1} \cdot r\, I = \frac{\beta}{\beta - 1} \cdot \frac{\sqrt{2r\gamma}}{AK^\alpha},$$

which is exactly (11). This derivation is fully consistent with [5], Chapter 5, where the exercise value is taken to be linear in the state variable; the result extends to the quadratic case via the sunk-cost equivalence. □ □

11 Conclusion

We have constructed a continuum economy in which the Cobb–Douglas production function $f(K) = AK^\alpha$ with quadratic adjustment costs $c(k) = \frac{\gamma}{2}k^2$ generates a class of best responses that are precisely real options: threshold rules with the option multiplier $\beta/(\beta - 1) > 1$ derived from smooth-pasting. Aggregation over normally distributed idiosyncratic shocks yields the scalar fixed-point equation (15). The main findings are:

- **Existence** follows from the IVT whenever A/γ exceeds a critical threshold.
- **Coordination failure** arises when $\alpha \approx 1$: the economy can settle at a low-activity trap (trivial equilibrium) or a high-activity boom, separated by an unstable tipping point.
- **Stability** is characterised by the derivative condition $|T'(K^*)| < 1$, with a quadratic Lyapunov function.
- **Structural GMM** identifies all six parameters from cross-sectional moment restrictions, with a sandwich asymptotic variance given by (25).
- **Algorithm 1** converges linearly at rate $\rho < 1$; Anderson acceleration achieves superlinear convergence.
- The framework is the scalar analogue of *deep equilibrium models*, with the IFT gradient (31) enabling bilevel optimisation.

Several extensions are left for future work: heterogeneous agents with persistent productivity differences; aggregate uncertainty of the kind studied in [9]; global-game refinements in the spirit of [8] for equilibrium selection; and continuous-time structural estimation of GBM parameters from high-frequency panel data on investment.

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Glossary

Anderson acceleration
An extrapolation technique that combines the last m iterates of a fixed-point iteration to accelerate convergence. Solves a constrained least-squares problem (28) to find the optimal convex combination; achieves superlinear convergence near stable fixed points [10, 1].

BIC
Bayesian Information Criterion. A penalised likelihood criterion $\ln N \cdot p - 2\ell$ (where p is the number of parameters) used for model selection. Consistently selects the true model as $N \to \infty$ under standard regularity conditions.

GBM
Geometric Brownian Motion. The SDE $\mathrm{d}X = \mu X\,\mathrm{d}t + \sigma X\,\mathrm{d}W$, whose log follows a Brownian motion with drift $\mu - \frac{1}{2}\sigma^2$ and variance σ^2. Standard model for productivity shocks and asset prices; see Assumption 2.1.

GMM
Generalised Method of Moments. A semi-parametric estimator that minimises a quadratic form in sample moment deviations (24). Encompasses OLS, IV, and MLE as special cases; the two-step variant uses an optimal weight matrix [6].

IFT *Implicit Function Theorem.* If $F(x, y) = 0$ with $F_y \neq 0$, then locally $y = g(x)$ with gradient $\nabla_x g = -F_y^{-1} F_x$. Applied here via (31) to differentiate the equilibrium $K^*(\boldsymbol{\psi})$ through the fixed-point constraint.

IVT
Intermediate Value Theorem. If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$, then $\exists\, c \in (a, b)$ with $f(c) = 0$. Used in Theorem 4.3 to establish interior equilibria.

NPV rule
Net Present Value rule. The prescription to invest whenever $V^{\mathrm{inv}}(\theta) \geq I$ (i.e. NPV ≥ 0). Under irreversibility, the true threshold exceeds the NPV trigger by the option multiplier $\beta/(\beta - 1) > 1$.

Option multiplier
The factor $\beta/(\beta - 1) > 1$ in equation (11). It embodies the value of the option to delay: greater volatility σ or lower drift μ_θ raises β, increases the multiplier, and defers optimal investment.

Partial expectation
$\Lambda(\theta^*) := \mathbb{E}[\theta\, \mathbf{1}_{\theta \geq \theta^*}]$, defined in equation (13). Summarises the contribution of active agents to aggregate capital; it is increasing in the mean μ and decreasing in the threshold θ^*.

Real option
The right (but not obligation) to undertake an irreversible investment. Analogous to a call option on a financial asset, but exercised on a real asset. The optimal exercise policy is the threshold rule (12).

Smooth pasting
The pair of boundary conditions (VM)–(SP) that characterise the free boundary θ^* in optimal stopping problems. Value matching equates levels; smooth pasting equates first derivatives, ensuring the value function is C^1.

SNE

Symmetric Nash Equilibrium. An aggregate capital level K^* satisfying (15): all agents use the same threshold $\theta^*(K^*)$ and aggregate demand exactly regenerates K^*.

Strategic complementarity

A property of payoffs such that $\partial^2\pi_i/\partial k_i \partial K > 0$: each agent's marginal return rises in aggregate activity. In this model, $\partial^2\pi_i/\partial k_i \partial K = \alpha A K^{\alpha-1}\theta_i > 0$ whenever $\alpha > 0$.

Tobit model

A censored regression model [6, Ch. 22]. Here applied with a structurally determined censoring point θ^*: agents with $\theta_i < \theta^*$ are observed at $k_i = 0$, entering the likelihood as left-censored observations; see (22).

The End